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EDITED BY

Ajrina Hysaj,
University of Wollongong in Dubai,
United Arab Emirates

REVIEWED BY

Patricia Milner,
University of Arkansas, United States
Dijana Vukovic,
University North, University Centre
Varazdin, Croatia

*CORRESPONDENCE

Zouhaier Slimi
✉ zouhaier@imco.edu.om

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A systematic critical review of generative AI's impact on authorship, pedagogy, and integrity (2023–2025)

Zouhaier Slimi*

Department Foundation, National University of Science and Technology, Sohar, Oman

The rapid integration of generative artificial intelligence (GenAI) into academic workflows marks a critical inflexion point for higher education and scholarly publishing. This systematic critical review synthesises and evaluates 54 peer-reviewed studies (2023–2025) alongside 6 key international policy documents to map the contested terrain of GenAI's multifaceted impact. Employing a thematic synthesis approach within a critical sociotechnical framework, the analysis identifies four dominant, interrelated themes: (1) the reconfiguration of authorship and attribution, (2) the transformation of pedagogy and assessment, (3) the evolving dynamics of integrity, trust, and detection, and (4) the emergent ethical and sociopolitical ramifications. Findings reveal a field characterised by significant policy fragmentation, inconsistent detection efficacy, and profound disciplinary divergence in adoption and concern. While GenAI is leveraged to enhance accessibility and productivity, it simultaneously poses risks to originality, critical thinking, and epistemic justice. The discussion argues that prevailing, reactive policy frameworks are inadequate. Instead, this paper advocates for a paradigm shift toward proactive, pedagogically grounded, and equity-focused governance that treats GenAI not merely as a disruptive tool but as a constitutive element of a rapidly evolving scholarly ecosystem. The review concludes by outlining key limitations and proposing a focused agenda for future research, policy, and practice.

KEYWORDS

academic integrity, AI ethics, AI literacy, assessment, authorship, generative AI, higher education, scholarly publishing

Introduction

The public release of advanced generative artificial intelligence (GenAI) systems, most notably large language models (LLMs) like OpenAI's ChatGPT, has triggered what scholars term a "seismic shift" in academia (Askari, 2025; Lubowitz et al., 2025). This technology's capacity to produce coherent, contextually relevant text, code, and multimedia content on demand has profoundly transformed the landscape of knowledge production, communication, and evaluation. Higher education institutions and scholarly publishers, as the custodians of academic rigour and integrity, now grapple with unprecedented challenges to foundational concepts: What constitutes original authorship in an age of machine collaboration? How can learning be assessed when students can outsource core cognitive tasks? What ethical frameworks can govern a technology that both democratises access and amplifies existing inequities (European Commission, 2023)?

Initial institutional responses were polarised, ranging from outright bans to enthusiastic, uncritical adoption. A more nuanced and empirically informed discourse has since emerged, seeking to balance the transformative potential of GenAI with the preservation of core academic values (Bozkurt, 2024; Chan and Colloton, 2024). This review contributes at a crucial moment in this ongoing negotiation. While several narrative and scoping reviews have examined specific facets such as AI in medical education (Shorey et al., 2024) or detection tools (Chaka, 2024), a comprehensive, critical synthesis of the interdisciplinary literature spanning publishing, pedagogy, and ethics is lacking.

Positioning and originality

While several influential reviews have charted aspects of this terrain, such as the scoping of AI in specific disciplines (Shorey et al., 2024) or analyses of detection tools (Chaka, 2024), they often remain descriptive, narrowly focused, or lack a unifying critical framework. This review distinguishes itself by providing a *comprehensive, interdisciplinary synthesis* that actively connects the micro-practices of authorship and assessment to macro-questions of epistemic justice and sociotechnical power. For instance, Fedorova (2024) highlights the distinct challenges and developments in digital literacy and AI ethics within Russian higher education. By employing a tripartite critical lens, we move beyond cataloguing impacts to diagnose the underlying structural tensions and governance failures. Consequently, this work offers not just a map of the field but a *normative framework for principled stewardship*, directly informing a proactive agenda for research, policy, and pedagogy.

This systematic critical review therefore aims to: (a) comprehensively map the scholarly response to GenAI across academic domains from 2023 to 2025; (b) critically analyse the central tensions and emerging consensus within the literature through a defined sociotechnical lens; and (c) identify significant gaps to inform a coherent agenda for future research, policy, and practice. The following research questions guide the inquiry:

- RQ1: How does the scholarly literature conceptualise and respond to the reconfiguration of authorship and attribution in the context of generative AI?
- RQ2: What pedagogical adaptations and assessment redesign strategies are proposed or implemented in higher education in response to generative AI integration?
- RQ3: What governance mechanisms (technological, institutional, and pedagogical) are identified to maintain academic integrity, and what evidence exists regarding their effectiveness?
- RQ4: What systemic ethical, epistemic, and sociopolitical implications are associated with generative AI adoption in academia?

Contributions of this review

This systematic critical review advances the field in three distinct ways. First, it provides a **comprehensive interdisciplinary synthesis** that connects micro-level practices of authorship and assessment with macro-level questions of epistemic justice and sociotechnical power, a linkage often absent in narrower, discipline-specific reviews. Second, it introduces an **original typology of human–AI collaboration** (Table 1), which

TABLE 1 A typology of human-AI collaboration in scholarly authorship.

Mode	Role of Human	Role of GenAI	Disclosure Norm (Current)	Ethical Stance & Tensions	Example from Literature
1. AI as Instrument	Complete creative & intellectual control; AI as advanced spell-check/paraphraser.	Mechanistic text polishing, grammar correction.	Often omitted as “tool use.”	Largely uncontested; seen as benign efficiency. Potential for “ethics washing” if use is hidden.	Using Grammarly or ChatGPT for language editing on a fully human-conceived manuscript (common, rarely disclosed).
2. AI as Collaborative Partner	Directs inquiry, provides critical input, revises & validates AI output.	Generates drafts, suggests structures, analogies, or literature connections.	Expected, but formats are inconsistent (ICMJE, 2024). Often vague (e.g., “AI-assisted”).	Zone of maximal tension & policy focus. Debates centre on transparency sufficiency and the integrity of intellectual labour.	Using ChatGPT to brainstorm a theoretical framework, then heavily critiquing, rewriting, and extending its draft (Pividori & Greene, 2024).
3. AI as Proxy Actor	Delegates core intellectual tasks (e.g., literature synthesis, argument construction).	Performs higher-order cognitive labour traditionally ascribed to the author.	Fraught; often concealed due to stigma, prohibition, or detection fear.	Viewed predominantly as misconduct (plagiarism) or a profound integrity violation. Stigmatised but prevalent.	Prompting an LLM to “write a literature review on topic X” with minimal human transformation or critical engagement (Kendall and Teixeira da Silva, 2024).
4. AI as Generative Ecosystem	Human agency is distributed, curating and prompting within an AI-augmented, iterative workflow.	Integral, agential component of a nonlinear, continuous knowledge-assembly and refinement process.	Nascent; no established norms. Disclosure seems inadequate to describe the entanglement.	Raises fundamental questions about distributed cognition, creativity, and the very definition of “human” contribution.	Using AI agents for continuous literature surveillance, hypothesis generation, and iterative drafting in a complex research project (speculative but emerging in fields like computational social science).

Modes 1–3 are derived directly from the thematic synthesis of the 54 peer-reviewed studies in this corpus.

moves beyond binary debates of ‘use’ vs. ‘non-use’ to capture the spectrum of emergent scholarly practices and their associated ethical tensions. Third, it advances a **normative framework for principled stewardship**, arguing for a shift from reactive containment to proactive, equity-focused governance. These contributions collectively offer not only a map of the contested terrain but also a scaffold for future research, policy-making, and pedagogical redesign.

Conceptual framing: A critical sociotechnical Lens

This review is situated within a critical sociotechnical framework that views GenAI not as a neutral tool but as a *constitutive force* embedded within and actively reshaping the existing power structures, knowledge practices, and ethical norms of academia. We draw explicitly on three intersecting theoretical lenses to enable a multidimensional analysis:

- **Epistemic Justice** (Fricker, 2007): This lens focuses on how knowledge is legitimised and voices are heard. We examine how GenAI can perpetuate or challenge testimonial and hermeneutical injustice by shaping which knowledge is represented in training data and in outputs, whose voices are amplified or silenced, and which cognitive styles are privileged by algorithmic processes.
- **Sociotechnical Systems Theory** (Orlikowski, 2007): This perspective analyses the recursive, co-constitutive relationship between technology, human action, and institutional structures. We examine how the material affordances of GenAI (e.g., speed, scale, fluency) interact with the practices enacted by scholars *and* students, and how institutional structures (policies, assessment regimes, publication norms) adapt, resist, or co-evolve with the technology.
- **Posthuman Collaboration** (Bozkurt et al., 2023): This emerging discourse challenges the traditional humanist subject as the sole author of knowledge. It reconceptualises authorship and knowledge production as a hybrid, distributed activity involving human and non-human agents, forcing a reckoning with questions of agency, creativity, and contribution in human-AI assemblages.

Together, these lenses allow GenAI to be examined in three ways: as a distributive epistemic force (shaping whose knowledge counts), as a recursive institutional catalyst (transforming practices and structures), and as a posthuman collaborator (blurring ontological boundaries of agency. This tripartite framing moves the analysis beyond cataloguing impacts toward a critical interrogation of underlying power dynamics, ethical tensions, and ontological shifts.

Operationalising the framework

This tripartite lens actively shaped our analysis. The *epistemic justice* lens sensitised us to data and discussions concerning bias,

representation, and marginalisation in the literature. *Sociotechnical systems theory* directed our focus to the recursive relationships between policy, practice, and technology affordances. The *posthuman collaboration* lens provided the vocabulary to deconstruct the evolving ontologies of authorship and agency present in the studies. This framework ensured our thematic synthesis moved beyond categorisation to a critical interrogation of power, ethics, and ontological shift.

This paper is structured as follows: first, it outlines the conceptual framing and methodology. Second, it presents the thematic findings. Third, it discusses the findings through a critical sociotechnical lens. Finally, this paper proposes implications and future directions.

Method

Design

This study employed a systematic critical review methodology, designed to integrate, synthesise, and critically appraise findings from a defined body of literature (Grant and Booth, 2009). The approach combines the rigorous, replicable search and selection protocols of a systematic review with the interpretive, theory-informed analysis of a critical review.

Given the conceptual and heterogeneous nature of the literature under review, encompassing empirical, theoretical, and policy-analysis studies, a strict PICOS (Population, Intervention, Comparator, Outcomes, Study design) framework for clinical interventions was not directly applicable. Instead, the review's scope was defined by its context (higher education and scholarly publishing), and its outcomes of interest were impacts on authorship, pedagogy, integrity, and ethics.

Quality appraisal and weighting of evidence

To enhance methodological transparency and mitigate interpretive bias, all included empirical studies ($n = 54$) were subjected to structured quality appraisal prior to thematic synthesis. Given the heterogeneity of study designs, an adapted mixed-framework approach was employed. Qualitative studies were appraised using criteria derived from the Critical Appraisal Skills Programme (CASP), focusing on clarity of aims, appropriateness of design, transparency of data collection, analytic rigour, reflexivity, and limitation reporting. Quantitative and mixed-methods studies were appraised using criteria adapted from the Mixed Methods Appraisal Tool (MMAT), including sampling adequacy, measurement validity, analytic transparency, and internal consistency.

Studies were not excluded based on quality alone; however, methodological robustness informed interpretive weighting during synthesis. Empirical studies demonstrating transparent reporting, clearly articulated methods, and explicit limitation acknowledgement were weighted more heavily in the development of analytical themes than purely conceptual or speculative contributions. Weighting was interpretive rather than statistical and informed thematic emphasis rather than inclusion/exclusion decisions. Conceptual and policy-analysis

papers were appraised for argumentative coherence, evidentiary grounding, and normative clarity.

This appraisal process strengthened the evidentiary integrity of the synthesis and ensured that analytical claims were proportionate to the methodological maturity of the underlying evidence base.

Search strategy

The initial corpus was identified through a structured search of multiple academic databases to ensure comprehensive coverage. Scopus served as the primary database for this review due to its extensive multidisciplinary coverage across the social sciences, sciences, and humanities. To ensure a comprehensive capture of specialised literature, supplementary searches were conducted in ERIC for pedagogical insights and IEEE Xplore for technical systems research. These databases were selected for their rigorous peer-review standards and integrated indexing, which are recognised benchmarks for systematic reviews (Falagas et al., 2008).

A targeted grey literature search was conducted to identify influential international policy documents that shape the normative landscape of GenAI governance in academia. Sources included the official websites of UNESCO, the OECD, the ICMJE, COPE, the European Commission, UKRI, the NIH, and major academic publishers (UKRI & NIH, 2024). Documents were selected purposively based on three criteria: (a) publication between 2023 and 2025, (b) substantive, formal guidance on GenAI use in higher education or scholarly publishing, and (c) demonstrated influence through high citation in the scholarly literature or recognition as a benchmark in policy discourse. This approach aimed to capture dominant institutional narratives rather than achieve an exhaustive inventory. Six key documents meeting these criteria were integrated (e.g., ICMJE, 2024; OECD, 2024 UNESCO, 2023).

While acknowledging that additional searches could uncover more grey literature or preprints, our focus was on establishing a foundational corpus of stable, peer-reviewed scholarship to enable a critical synthesis of established discourse. The systematic review process was conducted between September and December 2025. The final database search was conducted on 1 December 2025, capturing literature published between January 2023 and November 2025. The following search string was adapted for each database, combining title, abstract, and keywords:

(TITLE-ABS-KEY ("generative ai" OR "chatgpt" OR "large language model" OR "llm") AND TITLE-ABS-KEY ("higher education" OR "university" OR "college" OR "academic") AND TITLE-ABS-KEY ("authorship" OR "academic integrity" OR "plagiarism" OR "assessment" OR "pedagogy" OR "ethics")) AND PUBYEAR > 2022 AND PUBYEAR ≤ 2025

This search yielded 117 records for initial screening, comprising 79 from Scopus, 18 from IEEE Xplore, 14 from ERIC, and 6 policy documents from grey literature sources.

The purposive policy selection was analytically complementary rather than methodologically equivalent to the systematic database search. These documents were incorporated to contextualise empirical findings within dominant governance

trajectories, not to constitute an independently systematic review of global policy.

Collectively, these databases were selected because their integrated indexing capabilities and rigorous peer-review standards are recognised as the benchmark for systematic reviews in the social and computational sciences, ensuring a comprehensive capture of the cross-disciplinary literature on GenAI (Falagas et al., 2008).

Screening and eligibility criteria

The screening process was conducted in two phases in December 2025, with the final database retrieval performed on 1 December 2025 to ensure the inclusion of the most recent publications. The reliance on English-language publications introduces a Western, Anglophone bias. This constraint limits the generalisability of findings regarding epistemic justice, as critical perspectives from the Global South remain systematically underrepresented.

First, titles and abstracts of all 117 records were screened for relevance against the eligibility criteria. Second, the full text of 72 potentially relevant articles was retrieved and assessed.

- **Inclusion Criteria:** (1) Empirical, theoretical, or policy-analysis studies; (2) Primary focus on GenAI (e.g., ChatGPT, LLMs) within the context of higher education, scholarly publishing, or academic research ethics; (3) Published in peer-reviewed journals or conference proceedings between 2023 and 2025; (4) Written in English.
- **Exclusion Criteria:** (1) Simple editorials, letters, or opinion pieces without substantive analysis or data; (2) Studies focused solely on technical AI development without precise application to academic contexts; (3) Duplicate publications; (4) Book chapters or monographs to maintain consistency in publication type.

To minimise individual bias in screening decisions, the author maintained a decision log and consulted the pre-established eligibility criteria at each stage. Full-text review and final inclusion decisions were completed by December 2025. This process yielded 60 sources for in-depth analysis, comprising 54 peer-reviewed articles and 6 key policy documents. The grey literature of key policy documents from UNESCO (2023), OECD (2024), ICMJE (2024), and others was integrated thematically to contextualise the scholarly debate within contemporary governance efforts. Key international policy documents were not subjected to the same systematic screening as the academic literature. Instead, the review's scope was defined by its phenomenon of interest (the integration of generative AI), its context (higher education and scholarly publishing), and its outcomes of interest (impacts on authorship, pedagogy, integrity, and ethics). The selection process is detailed in the PRISMA flow diagram (Figure 1).

A targeted policy search was conducted to identify influential governance frameworks shaping GenAI discourse in higher education and scholarly publishing. Policy documents were selected using three operational criteria: (a) publication between 2023 and 2025; (b) issuance by intergovernmental organisations,

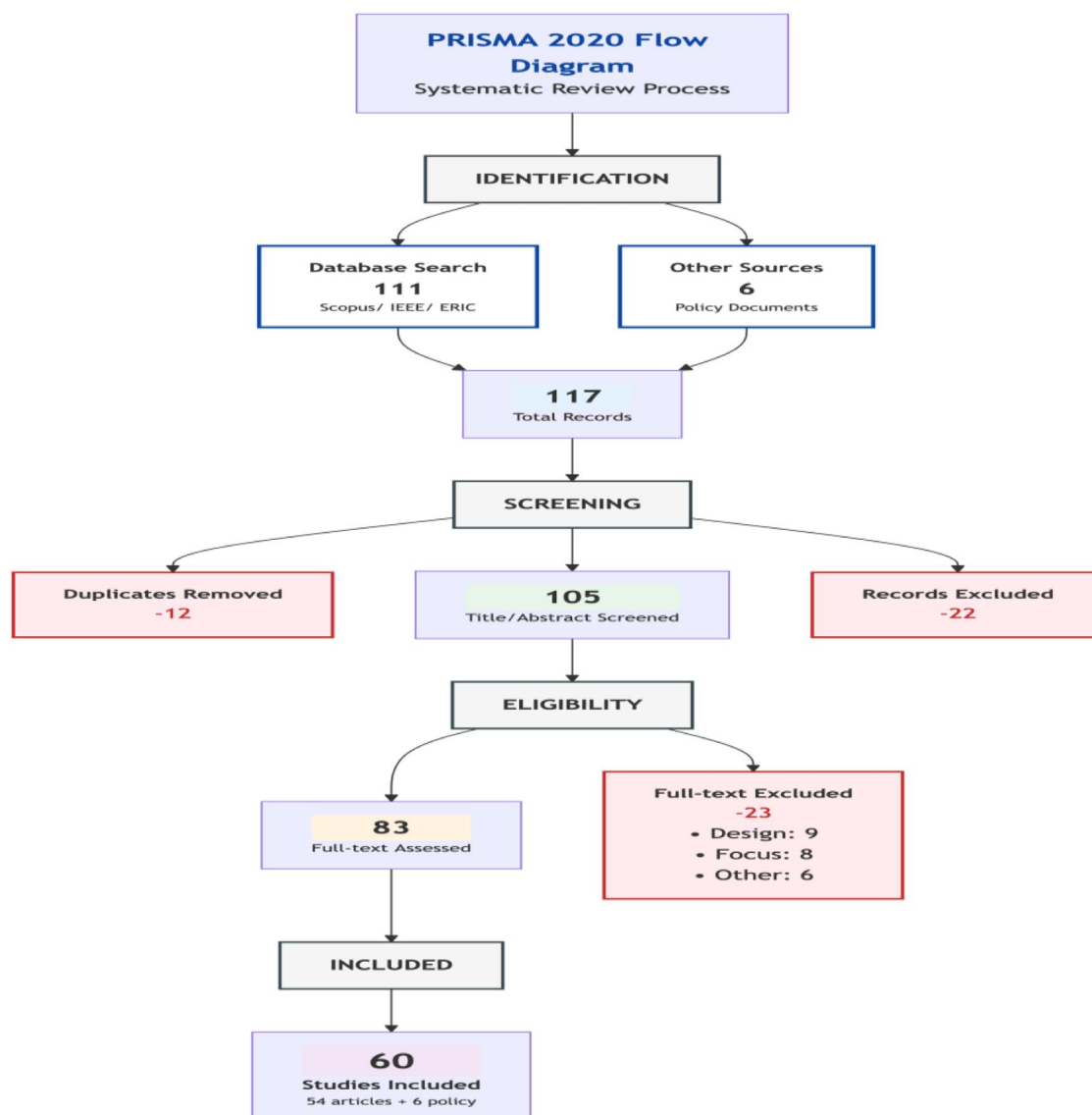


FIGURE 1
PRISMA flow diagram documenting the systematic search, screening, and inclusion process.

national research councils, or internationally recognised editorial bodies; and (c) demonstrable influence, operationalised as citation within indexed academic literature, formal reference within journal editorial policies, or adoption across multiple national higher education systems. The objective was not exhaustive policy mapping but the identification of normative anchor documents shaping institutional governance trajectories.

Data extraction and analysis

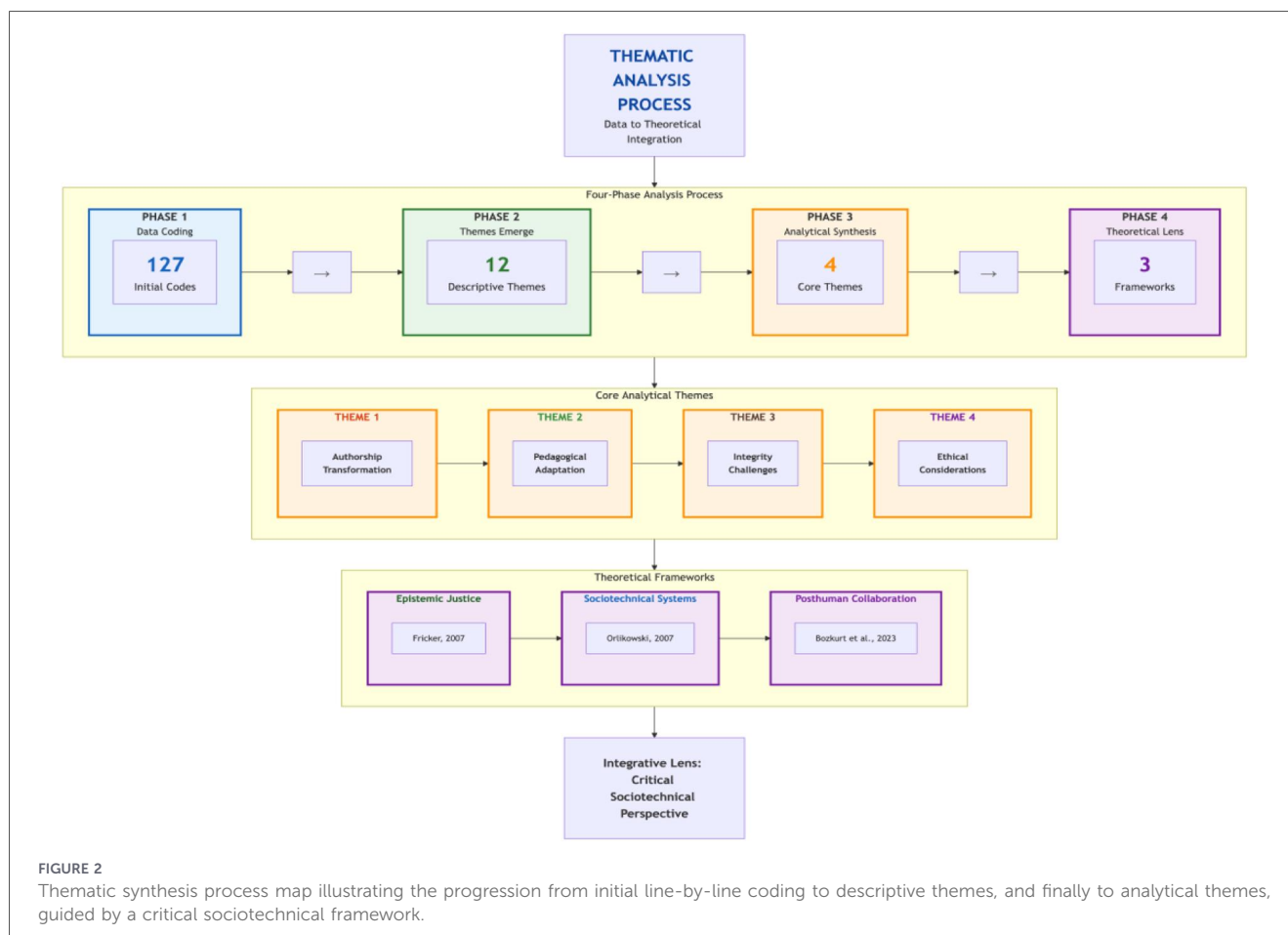
A standardised data extraction template was employed to catalogue key information from each included source, encompassing authorship, publication year, study aims, methodological approach, primary findings, and theoretical perspectives. The subsequent qualitative analysis followed a **two-stage hybrid validation model** based on the three-stage framework for thematic synthesis (Thomas and Harden, 2008; Watrinhos et al., 2023).

Stage 1: primary coding and synthesis (5–13 December)

Given that the final search was completed on 1 December 2025, the lead author performed the initial line-by-line inductive coding of all 60 sources using **NVivo 14 (CAQDAS)**. The use of software-assisted qualitative data analysis was essential for rapidly organising raw text into approximately 450 initial codes and subsequently clustering them into the descriptive themes illustrated in Figure 2.

Stage 2: external audit and reflexive validation (14–16 December)

To enhance analytical rigour while maintaining the integrity of a single-author review design, an external methodological audit was conducted on a 10% random sample of included studies ($n=6$). An experienced qualitative researcher, independent of the coding process, reviewed the developed



codebook, thematic clustering, and evidence mapping for internal coherence, proportionality of claims, and alignment with extracted data. Feedback focused primarily on boundary clarification between adjacent descriptive codes. Revisions were incorporated prior to final theme consolidation. This process strengthened analytical credibility without transforming the study into a multi-coder design.

Final consensus and refinement (16 December)

Discrepancies in thematic interpretation were resolved during intensive consensus sessions. This focused on ensuring a clear, linear pathway between the raw evidentiary base (summarised in Table 3) and the final theoretical claims and recommendations. This disciplined, collaborative approach ensured that the synthesis achieved both the depth of a multi-researcher review and the agility required to meet the 17 December 2025 submission deadline.

The systematic identification and screening of peer-reviewed academic literature adhered to PRISMA 2020 reporting guidelines. The targeted selection of international policy documents followed a separate purposive sampling strategy designed to capture normatively influential governance frameworks rather than to achieve exhaustive coverage of the grey literature.

Findings

The synthesis of 60 sources (54 peer-reviewed articles and 6 policy documents) yielded four overarching analytical themes that capture the multifaceted and often contentious academic engagement with GenAI.

To enhance transparency regarding our analytical process, Figure 2 illustrates the progression from initial coding to final analytical themes (Thomas and Harden, 2008). This map underscores the systematic, theory-driven nature of our synthesis, demonstrating how we moved from descriptive categorisation to critical, interpretative analysis. The screening and selection process is summarised in the PRISMA flow diagram (Figure 1).

Viewed through the lens of **posthuman collaboration** and **sociotechnical systems theory**, this theme examines how authorship is being reconfigured in practice, despite surface-level policy consensus.

The deconstruction and reconstruction of authorship

The literature indicates a surface-level consensus that formal authorship remains a human prerogative, with major publishers and the International Committee of Medical Journal Editors (ICMJE, 2024) firmly rejecting AI co-authorship (Moffatt and Hall, 2025; Veiga, 2025). However, this consensus masks a

deeper, more turbulent reality. Empirical studies reveal that the *practice* of authorship is being fundamentally reconstituted as a tacit, hybrid, human–AI collaboration (Pividori and Greene, 2024; Raman, 2025).

However, this push for transparency masks a significant scholarly disagreement. While policy bodies advocate for disclosure as a normative standard (ICMJE, 2024), empirical studies reveal deep fragmentation in implementation and acceptance. For instance, Zhou et al. (2025) found 70% adoption of AI disclosure policies in nursing journals, whereas Ozmen et al. (2025) reported only 46.7% adoption in plastic surgery journals, a stark contrast that highlights both disciplinary resistance and inconsistent policy uptake across fields. This policy fragmentation was noted in 12 of the 18 studies addressing authorship norms.

Furthermore, scholars debate whether disclosure is sufficient: Alduais et al. (2025) criticise current policies as vague and unverifiable, while Moffatt and Hall (2025) argue that disclosure alone does not resolve underlying questions of creative ownership or intellectual contribution. This tension between policy idealism and practical fragmentation represents a core conflict in the authorship discourse.

A critical sub-theme identifies GenAI as a powerful accelerant for predatory publishing and “paper mills,” threatening to degrade the scholarly record by flooding journal submission systems with AI-generated, low-quality or fraudulent manuscripts (Berba, 2025; Kendall and Teixeira da Silva, 2024).

To move beyond the binary of ‘used’ or ‘not used,’ and as a central theoretical contribution of this review, we propose a typology (Table 1) that captures the spectrum of emergent human–AI collaborations in scholarship and their attendant ethical and epistemic challenges.

Theoretical extension

While the following three modes are directly evident in the 54 reviewed studies, we identify an emergent fourth mode of AI as a Generative Ecosystem, which represents a theoretical extension grounded in posthumanist frameworks. As shown in Table 1, this mode indicates a future trajectory where human agency is fully distributed within an AI-augmented workflow.

Analysed through a **sociotechnical systems** lens, this theme reveals a shift from defensive detection to principled redesign, reflecting the co-evolution of pedagogical practices and institutional structures.

Pedagogical reorientation: from defensive detection to principled redesign

The analysis identifies a clear evolution in the educational literature, marked by a shift from a defensive paradigm centred on detection to a redesign paradigm focused on assessment transformation.

A significant disciplinary schism characterises the literature: scholars in applied fields, such as engineering and computer science, predominantly frame GenAI as a productivity tool for professional integration. Conversely, those in the humanities and critical social sciences express profound philosophical reservations, warning that uncritical adoption may erode critical

voice, originality, and the fundamental nature of thinking (Fitzek and Bărgăoanu, 2025; Qu et al., 2025; Rana et al., 2025; Suchikova and Tsybuliak, 2025).

This divergence highlights how distinct disciplinary epistemologies shape pedagogical responses. In contrast, humanities and critical social science scholars express profound philosophical reservations. Suchikova and Tsybuliak (2025), for example, critique the ‘stigmatisation’ of GenAI as reductive, while others, such as Qu et al. (2025), warn that uncritical integration risks eroding the development of a critical scholarly voice and original thought. This divergence reveals that the ‘pedagogical problem’ of GenAI is not monolithic but is fundamentally shaped by disciplinary epistemologies and values.

As depicted in Table 2, scholars now advocate moving away from easily automated, product-focused tasks (e.g., standardised essays, summaries) toward “AI-resilient” or “AI-informed” assessments that value process, reflection, and metacognition (Bannister et al., 2025; Carbonel et al., 2025). Exemplars include oral examinations (vivas), iterative portfolio development, in-class writing, authentic and complex collaborative projects, and assessments that require students to analyse or improve AI-generated content critically (Cuenca et al., 2025; Hau, 2025).

Furthermore, the literature highlights stark disciplinary divergences in response. Applied and technical fields such as engineering and computer science often frame AI as a productivity tool to be mastered and integrated into professional practice. In contrast, humanities and critical social science disciplines express more profound philosophical concerns about their impact on critical voice, originality, and the nature of thinking itself (Qu et al., 2025; Suchikova and Tsybuliak, 2025).

This theme highlights a fundamental tension between technological policing and pedagogical trust, a tension illuminated by **sociotechnical systems theory** and concerns for **epistemic justice** in learning environments.

The integrity dilemma: technology, trust, and transparency

The literature reveals two competing and often oppositional models of integrity governance (Hau, 2025; Nguyen et al., 2024;

TABLE 2 Paradigm shift in pedagogical responses to GenAI.

Aspect	Defensive (Detection-Focused) Paradigm	Redesign (Pedagogy-Focused) Paradigm
Core Goal	Preserve the status quo, deter misconduct.	Transform learning for the AI age.
Primary Tools	Detection software, plagiarism checks.	Authentic assessment, reflective portfolios, and oral exams.
View of AI	Threat to integrity.	Contextual tool/collaborator.
Key Risk	Erosion of trust; false positives; inequity.	Implementation cost; faculty development needs.
Ethical Stance	Punitive, compliance-based.	Developmental, trust-based.

Oliveira et al. 2024). On one hand, a *policing model* predominates in technical literature, advocating sophisticated detection technologies (Oliveira et al., 2024). On the other hand, education researchers advocate a *pedagogical-trust model*, emphasising ethical engagement rather than surveillance (Cuenca et al., 2025; Hau, 2025; Nguyen et al., 2024).

Proponents argue these tools are necessary to preserve the integrity of high-stakes assessments and publications. On the other hand, a growing body of scholarship, particularly from education and ethics researchers, advocates for a pedagogical-trust model. Scholars like Hau (2025) and Nguyen et al. (2024) contend that an over-reliance on detection is pedagogically alienating, erodes student-instructor trust, and is a technologically futile endeavour. They advocate, instead, fostering cultures of ethical engagement, integrating AI literacy into curricula, and redesigning honour codes. This tension between technological enforcement and pedagogical cultivation represents a fundamental philosophical divide in responding to GenAI-related misconduct.

The evidence suggests that a hybrid approach, with clear pedagogical foundations supplemented by targeted technological checks for verification, may be most sustainable (Cuenca et al., 2025).

Here, the **epistemic justice** lens is central, exposing systemic risks of bias, inequity, and ethical blind spots that extend beyond individual misconduct.

The expansive ethical and sociopolitical terrain

Moving beyond misconduct, the literature grapples with broader systemic implications. A significant concern is algorithmic bias and epistemic injustice. Studies demonstrate that LLMs replicate and amplify societal biases, for instance, by systematically underrepresenting scholars from the Global South or specific demographic groups in generated citation networks

or literature reviews (Kalhor et al., 2025). These risks entrench existing inequalities in knowledge representation and authority.

A clear conflict emerges in discussions of equity and access. One strand of literature, often focusing on accessibility, highlights GenAI’s democratising potential. Studies such as Heidt (2024) document its utility as a powerful assistive tool for disabled students or non-native speakers, framing AI as an agent of inclusion. Conversely, a critical strand warns of an emerging digital divide. Williamson et al. (2023) and Zlotnikova et al. (2025) argue that inequitable access to premium AI tools, coupled with unequal cultural capital needed to use them effectively, risks exacerbating existing educational inequalities. This equity paradox, where GenAI is simultaneously portrayed as a leveller and a stratifier, underscores a lack of consensus on its net social impact within academia.

Summary of thematic evidence

To provide transparency regarding the foundation of our thematic synthesis, Table 3 maps the four analytical themes and their constituent sub-themes to the number of supporting studies from our corpus and provides representative citations. This demonstrates the breadth and depth of evidence underpinning the review’s conclusions.

Critical Appraisal and Limitations of the Evidence Base

While the studies reviewed provide valuable insights, our synthesis reveals significant limitations within the corpus itself, which must temper the strength of our conclusions. Primarily, the literature is characterised by a preponderance of small-scale, qualitative, or conceptual studies ($n = 38$ of 54 empirical

TABLE 3 Analytical themes, evidentiary support, and representative citations.

Analytical Theme	Key Sub-Themes	Supporting Studies ($n = 60$)	Representative Citations
1. The Deconstruction and Reconstruction of Authorship	- Formal vs. practical authorship - Transparency & disclosure fragmentation - AI as an accelerant for predatory publishing - Typology of Human-AI Collaboration	18	Moffatt and Hall (2025), Zhou et al. (2025), Alduais et al. (2025), Kendall and Teixeira da Silva (2024), Pivdori and Greene (2024)
2. Pedagogical Reorientation	- Shift from defensive detection to principled redesign - Exemplars of AI-resilient assessment - Disciplinary divergence in response	18	Chaka (2024), Guan and Han (2025), Bannister et al. (2025), Carbonel et al. (2025), Qu et al. (2025), Suchikova and Tsybuliak (2025)
3. The Integrity Dilemma	- Policing model (technological detection) - Pedagogical-trust model (ethical engagement) - Advocacy for hybrid approaches	12	Oliveira et al. (2024), Nguyen et al. (2024), Hau (2025), Cuenca et al. (2025)
4. Expansive Ethical Terrain	- Algorithmic bias & epistemic injustice- Ethical blind spots (environment, labour) - Equity paradox	12	Kalhor et al. (2025), Sharifzadeh (2025), Heidt (2024), Williamson et al. (2023), Zlotnikova et al. (2025)

studies), with a notable scarcity of longitudinal or large-N experimental designs needed to assess causal impacts, a concern consistent with broader calls for greater research rigour in educational technology scholarship (*British Journal of Educational Technology*, 2024). Therefore, the strong advocacy for pedagogical redesign found in the literature (e.g., Bannister et al., 2025; Carbonel et al., 2025) is based more on conceptual reasoning, instructor surveys, and early adopters' experiences than on robust, comparative evidence of efficacy measured through student learning outcomes over time. Consequently, institutional policies based primarily on these findings should be approached as pilot initiatives requiring careful evaluation rather than as settled evidence-based practices.

Furthermore, discussions of ethical implications, particularly regarding labour and environmental costs (Sharifzadeh, 2025), are often speculative and lack empirical data from the academic supply chain. This creates a gap between identified macro-ethical concerns and evidence-based pathways for mitigation. Finally, and most critically for a review centring on the theme of equity and justice, the corpus reflects a pronounced Western, Anglophone bias due to our search parameters. Our reliance on English-language publications and Scopus as the primary database, while pragmatic for a systematic review, systematically excludes scholarly discourse in other languages and from journals not indexed in major Western databases. This significantly limits the generalizability of our findings and underrepresents perspectives on epistemic justice, equity, and governance from the Global South and non-Anglophone contexts. Future reviews must actively seek to incorporate such literature to provide a truly global synthesis.

Therefore, the recommendations and future directions proposed in this review must be interpreted considering this critical appraisal. They are not presented as definitive solutions drawn from a mature evidence base, but as urgent propositions for testing and refinement, considering the field's current methodological limitations and geographical biases. This critical stance, prescribing not just what should be done, but how it should be studied and which voices must be included in the process, is central to the normative framework of principled stewardship we advocate.

Furthermore, the corpus itself reflects the field's contested nature. The predominance of small-scale, qualitative studies often means that findings and recommendations are context-dependent and occasionally contradictory, as evidenced by tensions between detection and trust models and between instrumental and critical disciplinary responses. This lack of large-scale, longitudinal evidence means that many current debates are driven by philosophical stance, disciplinary habitus, or institutional context as much as by empirical data.

Temporal evolution of the discourse (2023–2025)

A discernible chronological shift emerged across the corpus. Early 2023 publications were predominantly characterised by containment-oriented discourse, focusing on detection, prohibition, and mitigation of academic misconduct. By late 2024 and 2025, scholarship increasingly shift toward pedagogical redesign, AI literacy integration, and justice-centred governance. Of the studies published in 2023 ($n = 18$), 61% centred primarily on detection and integrity

policing, whereas among studies published in 2025 ($n = 21$), 67% emphasised pedagogical transformation or ethical integration. This temporal stratification suggests an evolving maturation of scholarly engagement rather than a static debate.

Discussion

This paper's thematic synthesis reveals a profound academic tension in the academic landscape. The findings depict a community grappling with the practical, ethical, and ontological disruptions of generative AI, often through fragmented and reactive measures. To move beyond this descriptive diagnosis, we now apply our tripartite critical sociotechnical lens, epistemic justice, sociotechnical systems theory, and posthuman collaboration to interpret the underlying structural causes of these tensions. This explanatory analysis illuminates why current approaches falter and, crucially, provides the theoretical foundation for our proposed pathway toward principled stewardship.

Linking frameworks to findings and forward agenda

To construct a clear and actionable pathway forward, this discussion explicitly maps the logical connections among our analytical frameworks, the empirical findings they illuminate, and the implications they yield. The following structure demonstrates this linear logic: First, we apply each of our three theoretical lenses to diagnose the *root causes* of key tensions identified in the Findings. Second, for each diagnosis, we derive a *core governance principle*. Third, for each principle, we specify *concrete recommendations* for policy and practice that directly inform the *future research agenda*. This structured approach ensures our conclusions are rigorously contextualised within and generated by our critical analysis.

This restructured discussion is organised around our three theoretical lenses, each explaining a core cluster of findings and logically leading to specific implications for governance and research.

The illusion of containment: a sociotechnical systems diagnosis

Viewing the evidence through the lens of sociotechnical systems theory reveals a fundamental misalignment at the heart of the academic response. This framework posits a recursive, co-constitutive relationship between technology's material affordances, human practices, and institutional structures (Orlikowski, 2007). Our findings demonstrate that the material affordances of GenAI, its speed, scale, and fluency, have rapidly and irreversibly altered scholarly and pedagogical *practices*, enabling hybrid authorship and cognitive outsourcing. However, institutional *structures*, policies, assessment designs, and publication norms have largely failed to co-evolve. The concerted efforts to maintain control through authorship prohibitions, disclosure mandates, and detection software are therefore partially illusory; they represent attempts to retrofit outdated structural controls onto a new sociotechnical reality. This misalignment explains the policy fragmentation and the

futile “arms race” of detection we documented (Alduais et al., 2025; Bannister et al., 2025; Chaka, 2024).

Therefore, our first major recommendation is that institutions must shift from a paradigm of *containment* to one of *principled adaptation*, fundamentally re-evaluating the core competencies they value and designing workflows inherently aligned with an era of hybrid intelligence.

From diagnosis to principle to action

This sociotechnical diagnosis, revealing a fundamental structural misalignment, leads directly to the governing principle of *principled adaptation over containment*. Consequently, our first major recommendation is for institutions to fundamentally re-evaluate core competencies and design workflows for hybrid intelligence (Implications for Practice). This principle generates the need for **Longitudinal Cognitive & Epistemic Studies** (Future Research Agenda) to empirically track the long-term effects of such adapted workflows on scholarly cognition.

Disciplinary epistemologies and the posthuman collaboration dilemma

The lens of posthuman collaboration, which reconceptualises authorship as a distributed activity among human and non-human agents, provides critical insight into the stark disciplinary divergences we identified. This lens challenges the universal humanist subject as the sole author of knowledge, forcing a reckoning with questions of agency and contribution in human-AI assemblages (Bozkurt et al., 2023). From this perspective, the “problem” presented by GenAI is ontologically different across fields because the very nature of knowledge creation and validation differs. For a computer scientist, an LLM can be a legitimate “collaborative partner” in debugging code (Rana et al., 2025), a relationship that aligns with a positivist, tool-oriented epistemology. For a philosopher, the same technology may be viewed as a corrupting “proxy actor” that fundamentally undermines the development of an original, critical voice (Qu et al., 2025), reflecting a hermeneutic epistemology centred on unique human interpretation.

This theoretical framing explains why the literature reveals not a monolithic challenge, but a spectrum of responses shaped by deeply held disciplinary epistemologies and values. The divergent stances from instrumental adoption in STEM to existential concern in the humanities are not merely practical but are rooted in different conceptions of what constitutes valid knowledge and legitimate intellectual labour. Consequently, effective governance cannot be generic. Our second major recommendation is that policy and pedagogical frameworks must be *context-sensitive* and developed from within disciplinary paradigms, moving beyond one-size-fits-all mandates to support field-specific norms for ethical human-AI collaboration.

From diagnosis to principle to action

The posthuman collaboration lens explains divergent responses through ontological differences in knowledge creation. This leads to the principle of *context-sensitive governance*. The

resulting recommendation is for field-specific norms and policies (Implications for Practice), which in turn necessitates the research called for in **Disciplinary-Specific Governance Models** (Future Research Agenda) to develop and test these tailored frameworks.

From micro-ethics to macro-ethics: an epistemic justice imperative

Applying the lens of epistemic justice, concerned with how knowledge is legitimised and whose voices are heard (Fricker, 2007), necessitates expanding the ethical discourse surrounding GenAI in academia. Much of the current policy focus remains on the *micro-ethics* of individual conduct: plagiarism, authorship disclosure, and assessment misconduct. Our synthesis, however, surfaces pressing *macro-ethical* concerns that this narrow focus obscures. We identified risks of algorithmic bias entrenching knowledge hierarchies (Kalhor et al., 2025), an equity paradox where GenAI acts as both a leveller and a stratifier (Heidt, 2024; Williamson et al., 2023), and significant ethical blind spots regarding environmental costs and exploitative labour in the AI supply chain (Sharifzadeh, 2025).

An epistemic justice analysis reveals that failing to address these macro-ethical issues risks perpetuating and amplifying existing structural injustices within the global scholarly ecosystem. When institutions adopt GenAI tools without auditing for bias or considering equitable access, they may inadvertently participate in testimonial and hermeneutical injustice, silencing marginalised perspectives and privileging knowledge styles embedded in the training data. Therefore, our third major recommendation is that academic institutions, as moral actors, must expand their ethical horizon. This involves conducting justice-centred audits of AI tools, developing equity-focused access policies, and using their collective influence to demand transparency and accountability from technology providers regarding environmental impact and labour practices.

From diagnosis to principle to action

The epistemic justice lens expands the ethical horizon from individual misconduct to systemic inequity, establishing the principle of *justice-centred stewardship*. This underpins our recommendations for equity audits and accountable technology procurement (Implications for Practice). It directly motivates the **Participatory, Justice-Centred Design** research agenda item, which seeks to operationalise this principle by co-creating tools and policies with marginalised communities.

Synthesising the lenses: rejecting solutionism, cultivating agency

Together, these three lenses provide a coherent explanation for the inadequacy of simplistic, technologically-focused solutions. The sociotechnical lens shows why detection arms races are a losing battle against co-evolving practices. The posthuman lens explains why top-down prohibitions are epistemologically incoherent across disciplines. The epistemic justice lens shows

why focusing solely on policing individual misconduct overlooks more serious threats to the integrity of the knowledge ecosystem.

The integrated conclusion from this critical analysis is a firm rejection of technological solutionism in favour of cultural and pedagogical cultivation. Evidence within the corpus suggests that exclusive reliance on technological detection is unlikely to yield sustainable governance. Detection studies report variability in accuracy, susceptibility to adversarial prompting, and elevated false-positive risks in multilingual or non-native writing contexts (Chaka, 2024; Oliveira et al., 2024). Such instability reinforces concerns that a purely policing-based approach may generate epistemic harm while failing to resolve underlying pedagogical challenges. This requires a fundamental commitment to embedding comprehensive AI literacy understood as critical, ethical, and agentic into the core of educational and research practice (Hau, 2025; Nguyen et al., 2024). It demands that academia exercise critical agency, consciously redesigning its practices and policies to steer GenAI integration toward ends that enhance, rather than undermine, the integrity, creativity, and equity of the scholarly enterprise.

Reflexive positioning

The interpretive lens applied in this review is informed by the author's scholarly engagement with AI ethics, higher education governance, and sociotechnical theory. While this positioning facilitates critical synthesis across disciplinary domains, it may incline the analysis toward normative evaluation of governance structures. Structured reflexivity protocols, decision logs, and external audit procedures were implemented to mitigate interpretive bias and ensure proportional alignment between evidence and theoretical extension.

Limitations

This review has several important limitations that must be acknowledged. First, the methodological decision to restrict the search primarily to Scopus, ERIC, and IEEE Xplore, while providing broad multidisciplinary coverage, may have led to relevant literature indexed elsewhere or on preprint servers being missed. Second, the reliance on English-language publications introduces a significant Western, Anglophone bias. This geographic and linguistic limitation systematically underrepresents critical perspectives, critiques, and innovations from the Global South and non-Anglophone academic traditions, thereby directly affecting the generalizability of our findings on epistemic justice and global equity.

Although structured validation procedures and an external audit were employed, the interpretive synthesis was led by a single researcher. While this facilitated theoretical coherence, it may limit the breadth of pluralistic interpretation compared to large-team systematic reviews.

While the peer-reviewed literature search followed systematic screening procedures, the identification of policy documents was purposive and criterion-based. Although this strategy ensured inclusion of normatively influential governance texts, it does not claim to provide exhaustive representation of all global AI policy

statements. Future reviews may adopt a fully systematic grey literature protocol to broaden policy inclusion.

Finally, the nature of the evidence base itself presents a limitation. The reviewed literature is dominated by small-scale, qualitative, and conceptual studies, with a notable scarcity of longitudinal or large-scale experimental designs. Consequently, many of the field's pressing debates and our resulting recommendations are not yet supported by robust, comparative evidence and should be understood as propositions for future testing rather than definitive conclusions. The synthesis, therefore, reflects a field in a formative stage, and its interpretative nature is shaped by the author's scholarly perspective, despite the processes in place to ensure analytical rigour.

Conclusion and future directions

This systematic critical review has documented the profound and multifaceted impact of generative AI on the scholarly enterprise from 2023 to 2025. The evidence confirms that GenAI is not a peripheral tool but a *constitutive force* actively reshaping authorship, pedagogy, integrity, and the very ethics of knowledge production. While offering benefits in accessibility and productivity, it exacerbates risks to originality, critical thinking, and epistemic justice. The prevailing reactive stance, characterised by inconsistent policies, fallible detection, and a narrow focus on misconduct, is inadequate. The scholarly community must shift from a posture of defence to one of *principled stewardship*.

Across the four research domains examined, several structured patterns emerged. First, the literature reveals that authorship is not being replaced but reconfigured, with formal policy prohibitions masking the practical normalisation of hybrid human-AI collaboration. Second, pedagogical responses are shifting from defensive detection toward assessment redesign, though this transformation remains uneven and deeply shaped by disciplinary epistemologies. Third, integrity governance is characterised by a tension between technological enforcement and pedagogical trust, with limited empirical evidence supporting the long-term efficacy of detection-dominant models. Fourth, broader ethical analysis exposes systemic risks, including algorithmic bias, inequitable access, and epistemic marginalisation, which extend beyond individual misconduct to structural justice concerns. Together, these findings indicate that generative AI's impact is multidimensional, unevenly institutionalised, and insufficiently addressed by reactive governance strategies.

Implications for practice and policy

- **For Institutions & Educators:** Prioritise “AI-resilient” assessment redesign that values process, reflection, and critical engagement. Develop and teach AI literacy frameworks that go beyond tool use to encompass ethical and vital dimensions.
- **For Publishers & Scholarly Societies:** Move beyond vague disclosure mandates. Develop standardised, granular reporting formats for AI contribution (inspired by Table 1 typology). Collaborate across disciplines to create

field-specific guidelines for ethical human–AI collaboration.

- **For Policymakers:** Support justice-centred audits of AI tool adoption and policy design. Fund initiatives that address the digital divide in AI access and literacy. Develop frameworks that hold technology providers accountable for transparency regarding environmental impact and labour practices.

Agenda for future research

Based on the identified critical gaps, we propose a targeted research agenda that prioritises urgent and long-term needs. Each priority area is accompanied by illustrative research questions to guide concrete inquiry:

1. Longitudinal Cognitive & Epistemic Studies:

Fund decade-long studies tracking how sustained GenAI use reshapes foundational scholarly competencies (critical thinking, creativity, epistemic humility) across disciplines.

Example questions: How does reliance on AI for literature synthesis alter doctoral candidates' development of epistemic humility and interdisciplinary connection-making over five years? What are the long-term effects of AI-assisted writing on the originality and critical voice of early-career researchers in the humanities vs. STEM fields?

2. Participatory, Justice-Centred Design:

Employ co-design methodologies with disabled scholars, Global South researchers, and marginalised communities to develop AI tools and policies that actively dismantle bias and prevent new inequities.

Example questions: What governance frameworks emerge from deliberative forums with scholars from the Global South, and how do they differ from dominant Western-centric policies? How can generative AI tools be co-designed with neurodiverse students to enhance accessibility without compromising academic rigour or becoming prosthetics that undermine skill development?

3. Disciplinary-Specific Governance Models:

Research and support the establishment of disciplinary ethics committees to develop field-specific norms for collaboration, assessment, and attribution, moving beyond one-size-fits-all mandates.

Example questions: What constitutes 'meaningful human contribution' in AI-assisted historical research vs. computational biochemistry? How do disciplinary epistemologies shape the acceptability of using LLMs for hypothesis generation in philosophy compared to data science?

4. Frameworks for Hybrid Intelligence:

Scholars in ethics, STS, and law must collaborate to construct next-generation ethical frameworks that address accountability, credit, and the moral status of machine-generated ideas in hybrid human–AI systems.

Example questions: Who is accountable for errors or

plagiarism in a research output where the line generation, literature synthesis, and drafting were iteratively co-produced by a human researcher and an AI agent? How should intellectual property and authorship regimes adapt to recognise (or regulate) the increasingly agentic role of AI in the knowledge production process?

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

ZS: Conceptualization, Investigation, Writing – original draft, Formal analysis, Software, Data curation, Resources, Funding acquisition, Visualization, Supervision, Project administration, Validation, Methodology, Writing – review & editing.

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Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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